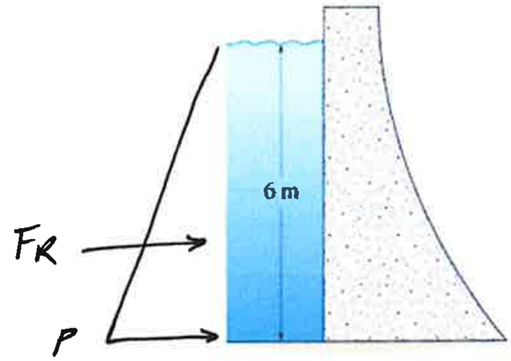
KEY

Practice Exam 4 from Dr. Koen

Practice Exam 4 – CE 2200, Dr. Koen

Section 1 – Fluid Pressure

1.1 Determine the **magnitude** of the resultant hydrostatic force acting on the dam. Width of the dam is 8 m. Density of water is 1.0 Mg/m³. $= 1000 \frac{kg}{m^3} \cdot 9.81 = 9810 \frac{N}{m^3}$



$= \gamma \cdot h = 6(9810)$

$= 58.86 \ kN/m^2$

$F_R = \frac{1}{2}bh = \frac{1}{2}(58.86)(6)(8)$

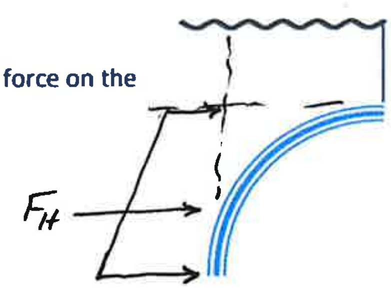
$\boxed{F_R = 1412.6 \ kN}$

1.2 Check all the **TRUE** statements about fluid pressure problems.

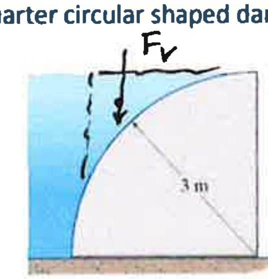
A. The center of pressure is always at the centroid of the submerged plate.
✓B. The resultant force passes thru the centroid of the loading diagram.
✓C. The center of pressure is the location where the resultant force strikes the plate.
D. The center of pressure is never at the centroid of the submerged plate.

1.3 What single shape would be used to determine the **Horizontal** component of force on the side of the curved constant width plate submerged in fluid.

Trapezoid



1.4 Determine the **Vertical** force component that the water exerts on the side of the quarter circular shaped dam **per meter of length**. Density of water is 1.0 Mg/m³.



$F_V = VOL \cdot \gamma$

$VOL = \left[3 \cdot 3 - \frac{\pi}{4}(3)^2\right](1m)$

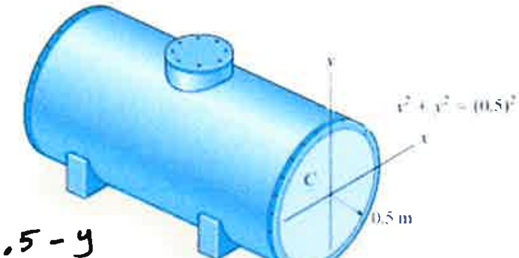
$= 1.931 \ m^3$

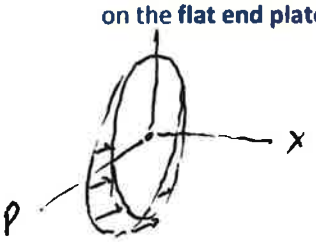
$= 1.931 \left(9810 \frac{kN}{m^3}\right)$

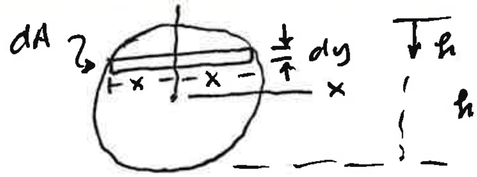
$\boxed{F_V = 18.95 \ kN}$

Integral on TI89: $\int((.5-y)*\sqrt{.25-y\hat{}2}), y, -.5, .5) = .19635$

1.5 Determine the **magnitude** of the resultant hydrostatic force acting on the **flat end plate "C"** of the tank. Density of water if 1.0 Mg/m³.







$h = .5 - y$

$p = \gamma h = 9810(.5-y)$

$x^2 = .25 - y^2$

$x = \sqrt{.25-y^2}$

$dA = 2x \cdot dy$

$= 2\sqrt{.25-y^2}\,dy$

$dp = p \cdot dA = 9810(.5-y)(2)\sqrt{\quad}\,dy$

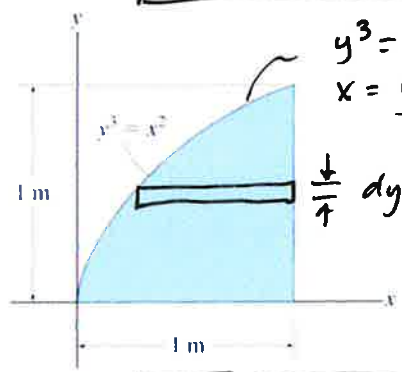
$F_R = \int_{-.5}^{.5} 9810(2)(.5-y)\sqrt{.25-y^2}\,dy$

$= 9810(2)(.19635)$

$\boxed{F_R = 3.85\ kN}$

Section 2 – Moment of Inertia

2.1 Determine the moment of inertia for the shaded about the x axis.



$y^3 = x^2$

$x = y^{3/2}$

$I_x = \int_A \tilde{y}^2 dA$

$dA = (1-x)\,dy$

$= (1-y^{3/2})\,dy$

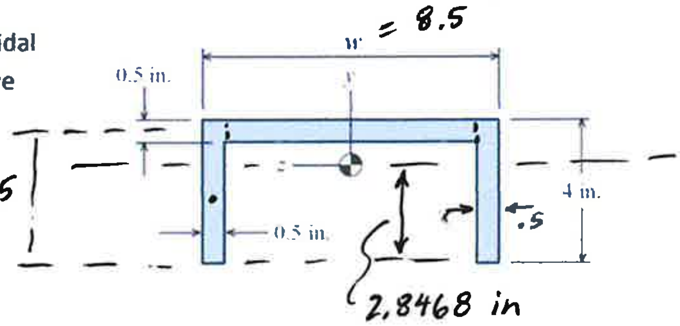
$= \int_0^1 y^2(1-y^{3/2})\,dy$

$= \int_0^1 (y^2 - y^{7/2})\,dy = \left(\frac{1}{3}y^3 - \frac{2}{9}y^{9/2}\right)_0^1 = \frac{1}{3} - \frac{2}{9} = \boxed{\frac{1}{9}} = \boxed{0.111\ m^4 = I_x}$

2.2 TRUE FALSE For a general area, the parallel axis theorem would be used when calculating Iy and using a horizontal differential elment. ~ ⊥ to y axis.

$3.75 - 2.847 = .903''$

2.3 If *w* = 8.50 in., find the moment of inertia about the centroidal *z*-axis. The centroid of the shape is located 2.8468 inches above the bottom of the shape.



$I_z = \Sigma(\bar{I} + Ad^2) =$

$= \left[\frac{1}{12}(7.5)(.5)^3 + 7.5(.5)(.903)^2\right]$

$+ 2\left[\frac{1}{12}(.5)(4)^3 + 4(.5)(.847)^2\right]$

$I_z = 3.136 + 2(4.1015)$

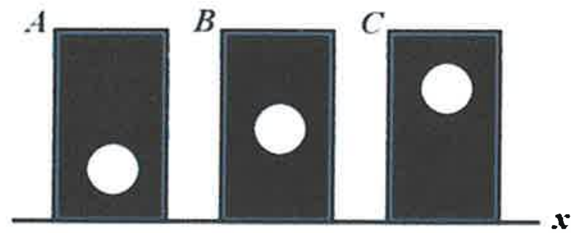
$\boxed{I_z = 11.34\ in^4}$

2.4 A general shape with an area of 20 in^2 has a moment of inertia of 106.7 in^4 with respect to the x axis. The centroid is located 2 inches above the x axis. What is the moment of inertia with respect to a centroidal x' axis?

$$I_x = \bar{I} + A d_y^2$$

$$\bar{I} = I_x - A y^2 = 106.7 - (20)(2)^2 \qquad \boxed{\bar{I} = 26.7 \text{ in}^4}$$

2.5 The figure shows three areas. Each area is a rectangle with a missing circle. The three rectangles have the same dimensions, and the three missing circles have the same dimensions. Which area has the largest moment of inertia about the *x* axis?



$$I_x = \int_A y^2 \, dA$$

I_x for A is largest.

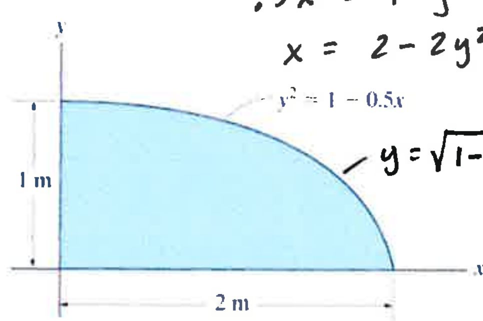
Section 3 – Product of Inertia

3.1 When using the parallel axis theorem to determine the product of inertia for an area about a set of non-centroidal x-y axes, $I_{xy} = \bar{I}_{x'y'} + A d_x d_y$, the second term on the right side of the equation is:

A. Area times the differential widths dx and dy
✓B. Area times the distances between the x-y origin and the x'-y' origin.
C. Always positive
D. Always negative
E. Always zero

$$.5x = 1 - y^2$$
$$x = 2 - 2y^2$$

3.2 Determine the product of inertia of the shaded area with respect to the *x* and *y* axes.



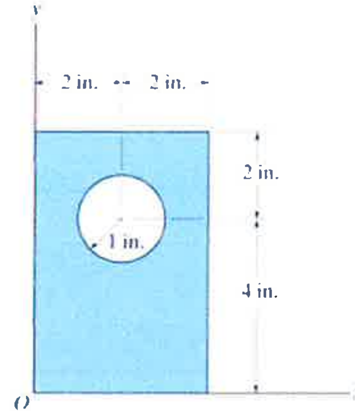
$y = \sqrt{1 - .5x}$

$$I_{xy} = \int_0^2 \int_0^{\sqrt{1-.5x}} xy \, dy \, dx$$

TI89 : $\int(\int(x*y, y, 0, \sqrt{1-.5*x}), x, 0, 2)$

$$= \boxed{.333 \text{ m}^4 = I_{xy}}$$

3.3 Determine the product of inertia for the shaded area with respect to the x and y axes.



$$I_{xy} = \Sigma(\bar{I} + A d_x d_y)$$

$$= [0 + (4)(6)(2)(3)]_{\square}$$

$$- [0 + \pi(1)^2(2)(4)]_{\circ}$$

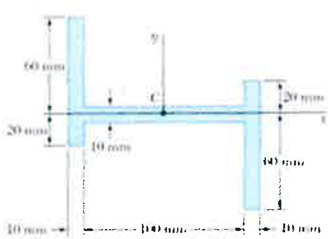
$$\boxed{I_{xy} = 118.9 \text{ in}^4}$$

By inspection, determine the sign of Ixy for figures based on the coordinate systems given. Choose from the following choices: positive negative zero

3.4

$-dx$

$+dy$

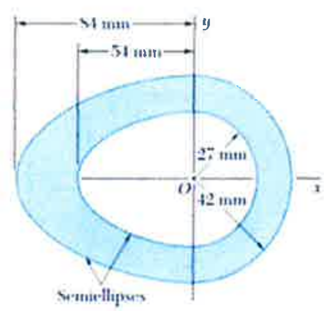


$-dy$

$+dx$

NEG

3.5

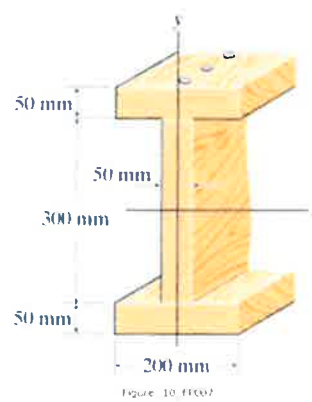


ZERO

(symmetric about x and y)

Section 4 – Inclined Axis and Principle Moments of Inertia

4.1 The orientation (*relative to the xy axes shown*) of the **principal axes** with origin at the centroid for the cross sectional area would be ______ degrees.



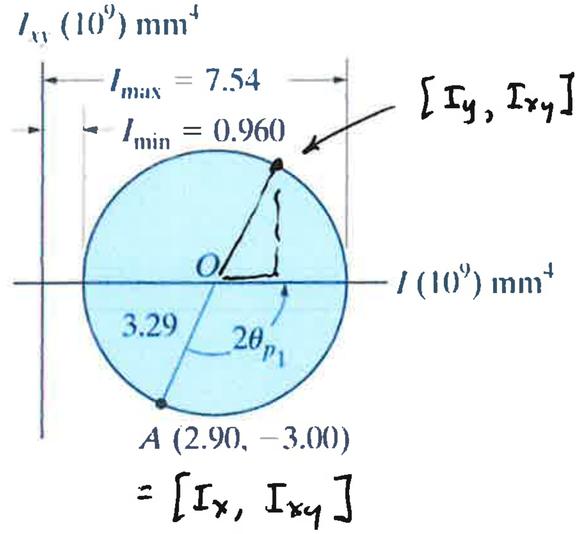
zero degrees.

$I_{xy} = 0$ since symmetric, and if $I_{xy} = 0$, then this must be a principal orientation.

$I_{xy} = 0$

A **Mohr's circle** has the reference point **A (Ix, Ixy)**.

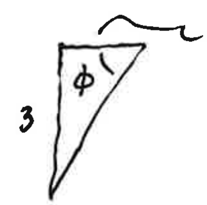
The **center** of the circle is at **4.25** and the **radius** is **3.29**



4.2 Iy = 5.6 $10^9 mm^4$

4.3 I_{min} = .96 $10^9 mm^4$

4.4 2θ = 114.2°.



$\Delta y = 4.25 - 2.9$
$= 1.35$

$\phi = \tan^{-1} \frac{3}{1.35}$
$= 65.8°$

$I_y = 4.25 + 1.35$
$= 5.6$

$2\theta = 180 - 65.8 = 114.2°$

4.5 On the **actual cross section**, the **major** principal axis would be shown rotated 57.1 degrees CCW from the positive x axis.

$$\theta_{P1} = \frac{114.2}{2} = 57.1°$$

Answers	
1.1 1413 kN	3.1 B
1.2 B and C are true	3.2 0.333 m^4
1.3 Trapezoid	3.3 119 in^4
1.4 18.9 kN	3.4 negative
1.5 3.85 kN	3.5 zero
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2.1 0.111 m^4	4.1 0 degrees
2.2 TRUE	4.2 5.6
2.3 11.34 in^4	4.3 .960
2.4 26.7 in^4	4.4 114.2 degrees
2.5 A	4.5 57.1 degrees CCW